Unified Feature Store

AI Platform

ebay

Agenda

- Feature Pain Points in Production AI
- Unified Feature Store @ AIP
- Feature Store Key Metrics
- Useful Links

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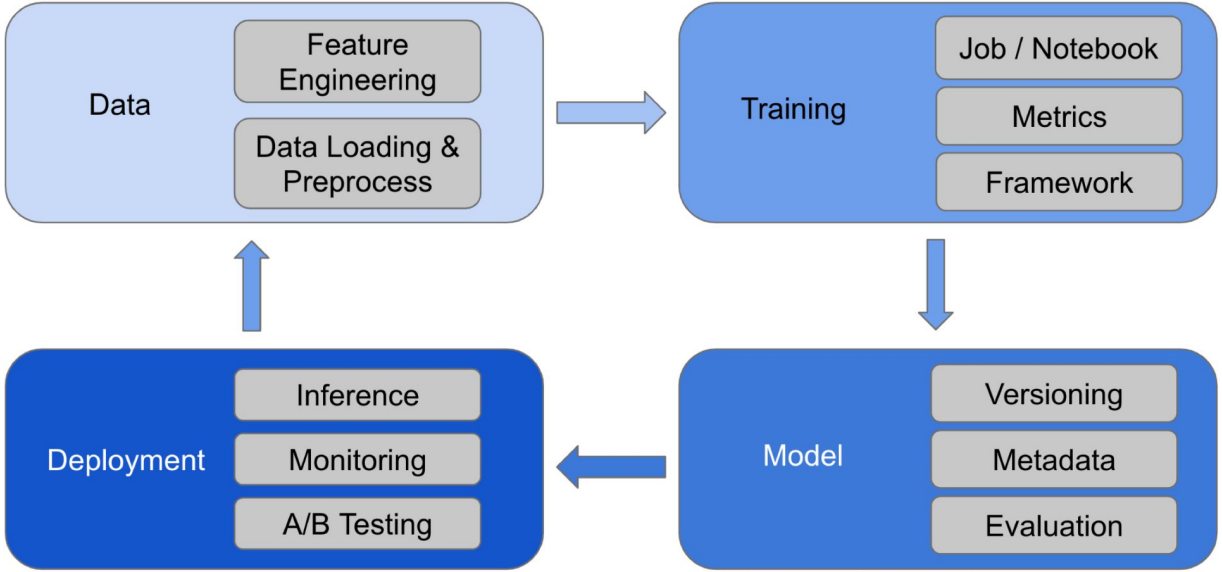
AI Project Life Cycle



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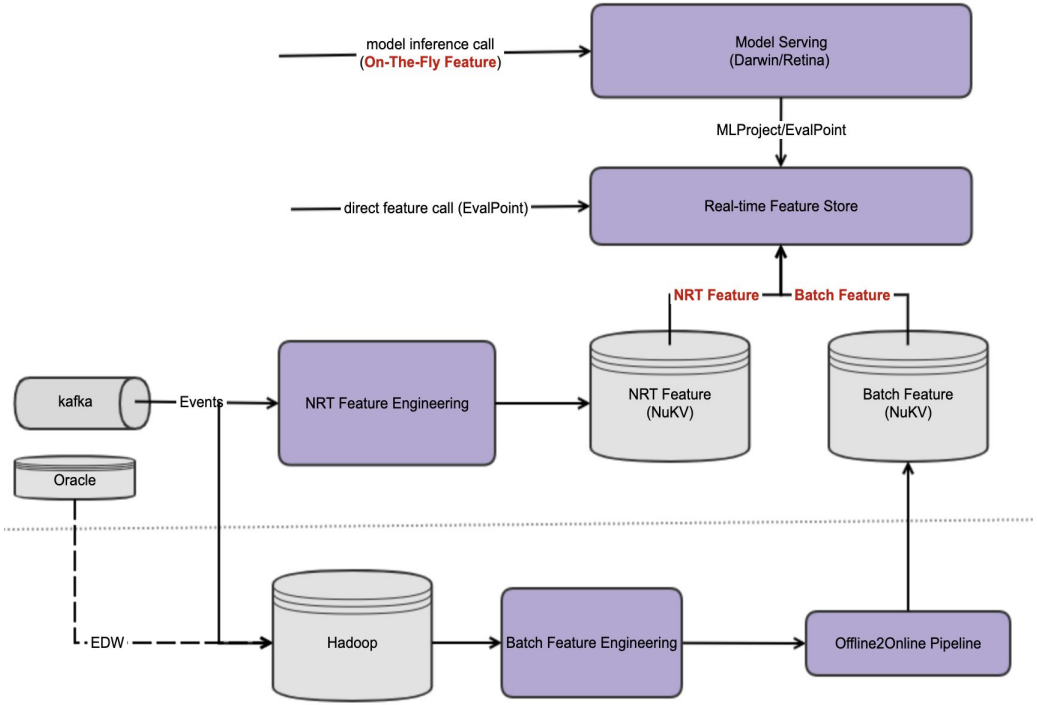
Feature 101

Feature Types

- **On-The-Fly (OTF) Feature**
  - short latency (ms)
  - Context data or real-time DB/Service query
- **Near-Real-Time (NRT) Feature**
  - Async process (seconds)
  - Streaming events.
- **Batch Feature**
  - Batch process (hours or days)
  - Data warehouse tables

Feature Storage

- Online: NuKV, latest value, serving in real time application
- Offline: HBase/HDFS, point-in-time value, feature simulation for research & training



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Major Pain Points in Feature

Long feature TTM (Time-to-Market)

- Lack of point-in-time online/offline feature parity
- Communication cost cross teams is not cheap
- Lack of self-service and automation tools for feature life-cycle management and integration

Feature discovery & reuse

- Need feature catalog for discovery and exploration
- Need feature data lineage information
- Lack of unified loading key management cross all use cases

Feature data quality monitoring & alerting

- Need active monitoring & alerting on feature data validation and data shift

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Architecture Visions & Capabilities

- Enable **self service** for feature engineering **with governance**
- Leverage **unified DSL** to define features for both online and offline
- Manage the whole **feature life cycle**
- Support seamless fast feature **deployment & release**
- Enable online/offline **data parity,** support feature point-in-time & backfill
- Provide **feature catalog** for discovery and reuse cross eBay

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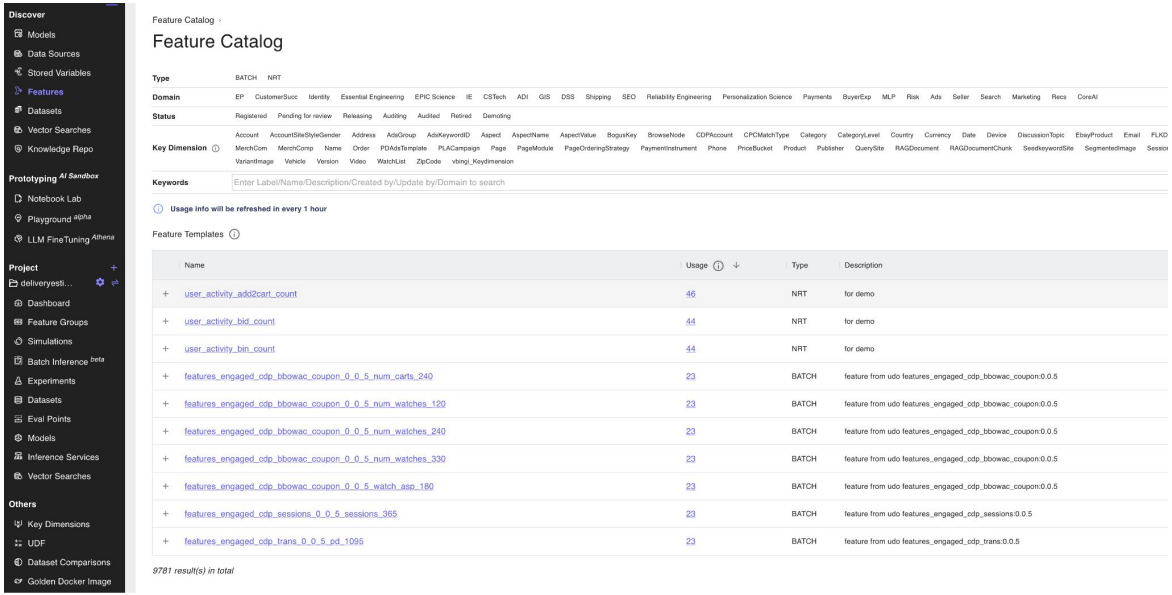
Unified Feature Store

- Feature versioning
- Feature lifecycle management
- Feature consistency
- Feature discovery and reuse
- Online/offline feature parity
- Fast feature backfill
- Feature access control
- Feature documentation & analysis



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Feature Store in Industry

- Uber: Michelangelo Palette
- Airbnb: zipline A Declarative Feature Engineering Framework
- Google: Vertex Feature Store
- AWS: SageMaker Feature Store
- Databricks: https://www.databricks.com/product/feature-store
- Feast/Tecton: https://github.com/feast-dev/feast
- Feathr: https://github.com/linkedin/feathr

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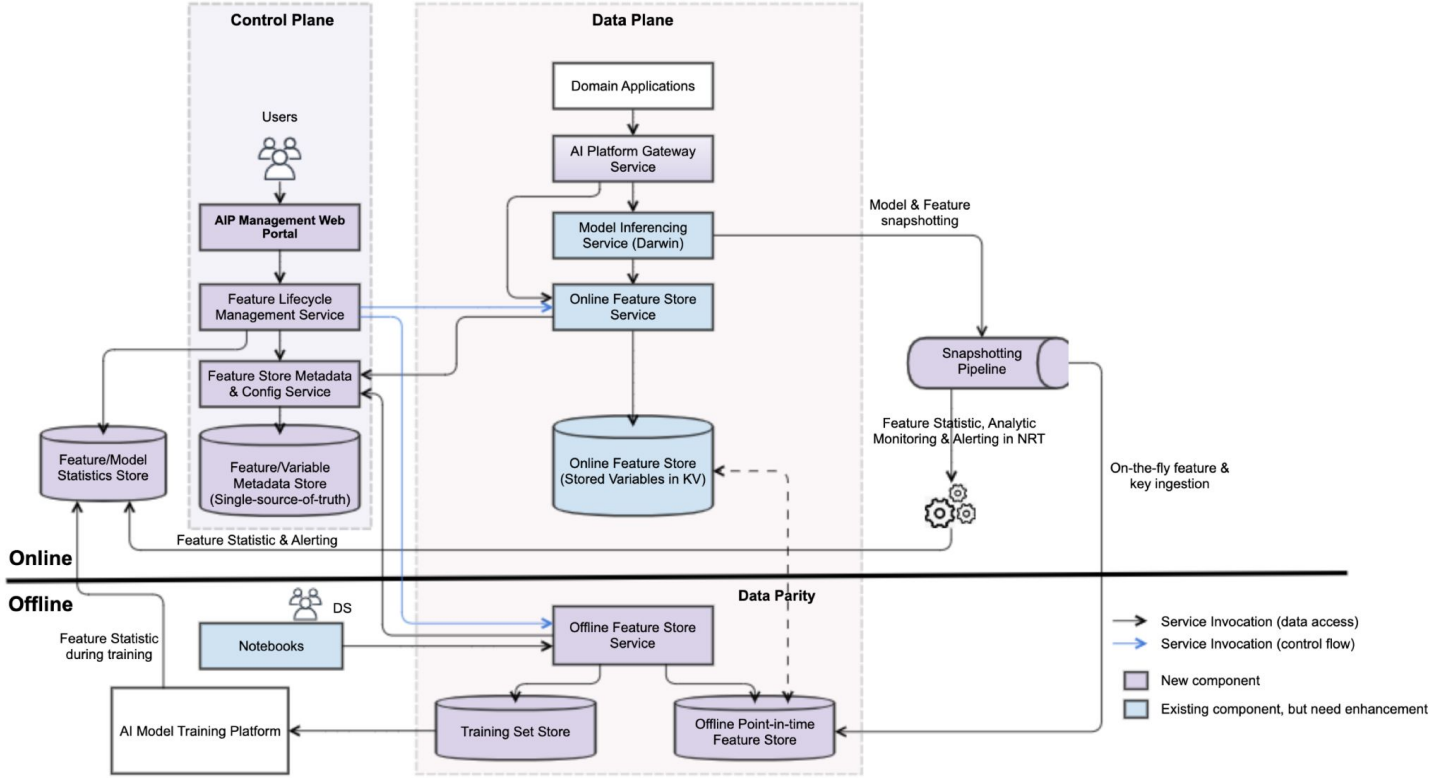
Unified Feature Store Architecture



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DSL Based Feature Template Definition

- Strong data types
- Interoperable with Java
- Static validation and dependency extraction
- UDF extension
- High execution efficiency

```
define variable gcmcnto_transshipaddr_acct_transip(KeyDim.Address address, long pit) as long {
    metadata : {
        "description" : "Count of linked ip by traverse path transaction shipping address -> a
    };

    local udo = @harvestLoad(address, "rgf_gvar_cmcnt_shipaddr_usr_ip", "0.1.0", pit);
    udo.gcmcnto_ip;
};
```

SW reading computation DSL

```
define variable sum_purchase_order_usd_by_shipaddr_sw(KeyDim.Address key, string timeWindow, long pit = 0) as double {
  metadata: {
    "import": ["com.ebay.nrt.da.rollup.AggFuncEnum"],
    "@param_timeWindow": "24h"
  };

  local swVar = @swLoad(key, "purchase_order_by_shipaddr_sw");
  if (swVar == null) {
    0;
  } else {
    local result = swVar.aggregate(pit, ::AggFuncEnum.SUM, timeWindow);
    if (result == null) {
      0;
    } else {
      result;
    }
  }
}
```


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Key Space Management

- Manage all the loading keys
- Each key should have its unique name, belong to a key dimension
- Provide catalog for key discovery and life-cycle management

General Dimensions

ImageGuid

ImageURL

Domain Dimension

AIP Account

AIP Address

AIP Date

AIP Order

AIP Video

Account

Key Dimension Name

Account

Data Type

STRING

Domain

Marketing Tech

Key Name

BuyerId

SellerId

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Feature Definition


```
{
    "id": "0001e8fc-2f89-4a12-a43a-01e26e494f86",
    "name": "gcmcnto_transshipaddr_acct_transip:BuyerAddress",
    "readingKey": [
        "BuyerAddress"
    ],
    "featureTemplate": "gcmcnto_transshipaddr_acct_transip"
}
```

```
{
    "id": "029755c6-b4f1-4d3c-b653-e56da7a5b4b3",
    "name": "sum_purchase_order_usd_by_shipaddr_sw^2h:BuyerAddress",
    "featureTemplate": "sum_purchase_order_usd_by_shipaddr_sw",
    "readingKey": [
        "BuyerAddress"
    ],
    "window": "2h"
}
```


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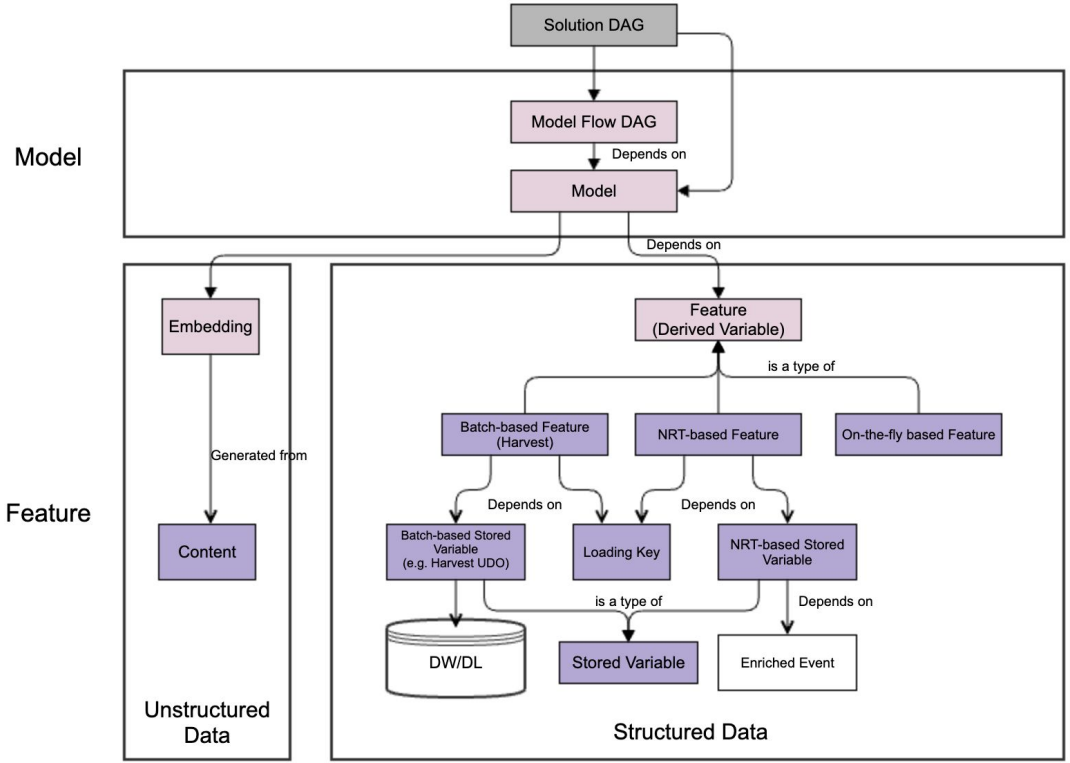
Feature/Variable Entity Dependency Model and Terminology



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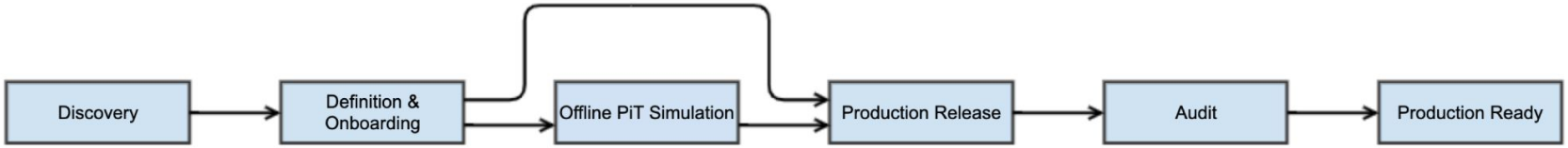
Feature Life Cycle Management



Platform Capabilities

- Entity catalog
- Entity discovery & search
- Data lineage extraction

Platform Capabilities

- Standardized entity onboarding
- Standardized metadata capturing

Platform Capabilities

- Feature offline PiT Simulation

Platform Capabilities

- Static validation
- Fast analytic release pipeline
- Fast backfill

Platform Capabilities

- Feature value snapshotting and validation
- Match rate report against PiT simulation

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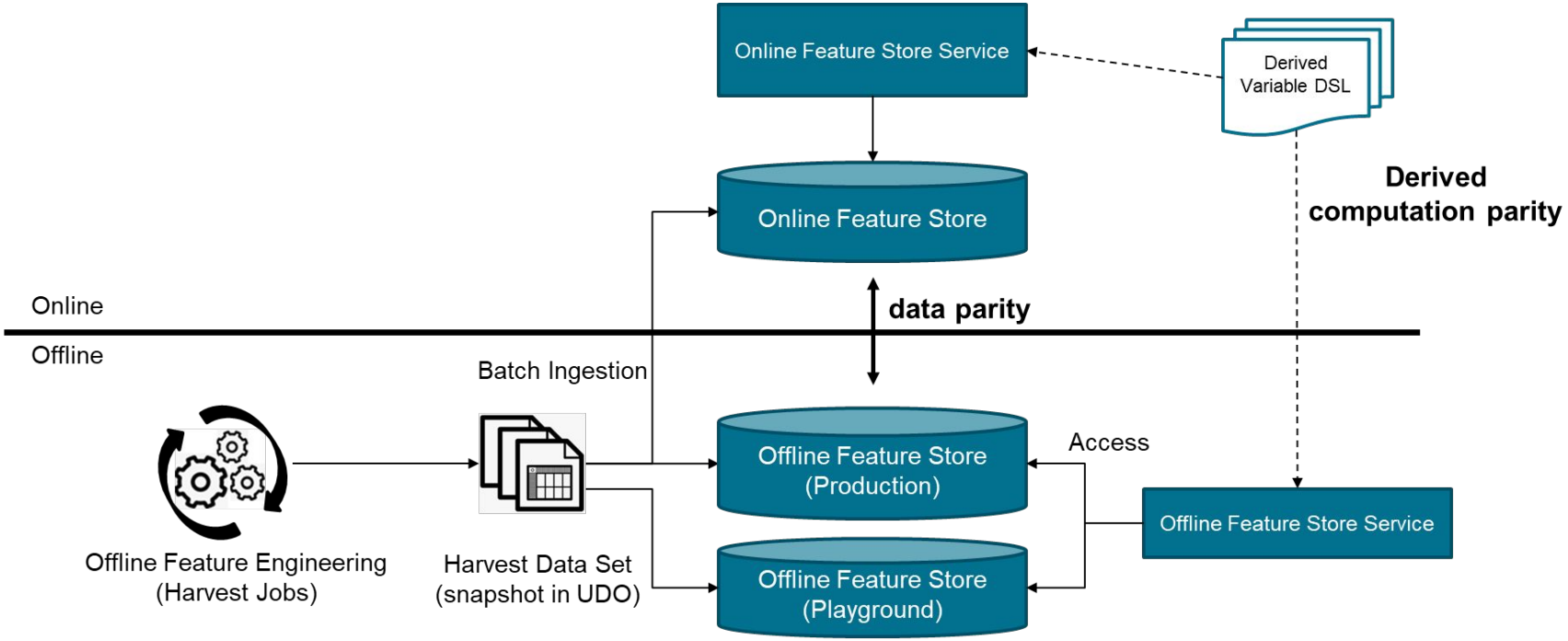
Batch Feature



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Batch Feature Onboarding

* Type ○ Event ◉ Batch

* Publish Strategy ○ Incremental ⓘ ○ Snapshot ⓘ

* HDFS Location /apps/b_hrvst/prod/release/parquet/b_mlops/mlops_sc_item_rec_latest_udo/prod [Load]

This approach only support < 100GB data storage for single case. Please consult #ml-platform-support if you have larger data.

Source Schema

Search...

Field Name	Field Type	Array Element ...	Is Primary Key	Key Dimension	Read Key
segmented_image_id	STRING		False		
site_id	INTEGER		False		
recs	STRING		False		

3 result(s) in total

1

Search... Columns Filters Density Export

segmented_image_id	site_id	recs
MzhkMzVjMjgtZjZiMi00M2U3LTgyZjUtZjIxMDQyYjk4OTRlXzM=	0	{"groupName":"38d35c28-f6b2-43e7-82f5-f21042b9894e_3","rankedItems":[{"summary":{"itemId":325962917679,"itemVrtId":0}
YzY3YTRiNWEtNDU4MS00MmE0LWE4QTMtZGFjMmRhY2JkOTIyXzQ=	0	{"groupName":"c67a4b5a-4581-42a4-a893-dac2dacbd922_4","rankedItems":[{"summary":{"itemId":254461610930,"itemVrtId":
dHJpdG9uX2ZsdXhlMjAxXzA=	0	{"groupName":"triton_flux_201_0","rankedItems":[{"summary":{"itemId":305080057416,"itemVrtId":604047625856},"modelInfo"
YWMzMjM3ZDEtMDk0Yy00NGUwLTg3OTEtZWY3NDBiNjEyNDExXzI=	0	{"groupName":"ac3237d1-094c-44e0-8791-ef740b612411_2","rankedItems":[{"summary":{"itemId":265104979559,"itemVrtId":0
OTNjYzNjZDEtNDVhZS00YmMyLTg1MDktNmNkZTgwMzEwMWViXzM=	0	{"groupName":"93cc3cd1-45ae-4bc2-8509-6cde803101eb_3","rankedItems":[{"summary":{"itemId":395109791881,"itemVrtId":
MzE4YjViOWUtNmE4NC00OGQyLThhZGEtYjc3OGU2YTJiOTVlXzE=	0	{"groupName":"318b5b9e-6a84-48d2-8ada-b778e6a2b95e_1","rankedItems":[{"summary":{"itemId":305259718377,"itemVrtId":
YmVmOWNiZWQtZmZhMi00ZGQzLWE4NTYtYTM2YTFlN2RiNjE3XzE=	0	{"groupName":"bef9cbed-ffa2-4dd3-a856-a36a1e7db617_1","rankedItems":[{"summary":{"itemId":196568529426,"itemVrtId":4
NThjOGE4YzMtYzA1ZS00MGQ4LTk0NWMtMTViMmJkMWUyOTliXzM=	0	{"groupName":"58c8a8c3-c05e-40d8-945c-15b2bd1e299b_3","rankedItems":[{"summary":{"itemId":226448996299,"itemVrtId":
ZGY5YWUyNzYtNzgwYy00NDE1LTg1ZTUtNTk2NWEzMGViNWU0XzM=	0	{"groupName":"df9ae276-780c-4415-85e5-5965a30eb5e4_3","rankedItems":[{"summary":{"itemId":146116563489,"itemVrtId":
NzFjYjIyNGUtYmVkMi00NDY3LWJlNWYtNWRlNjk4YmZjZTZkXzA=	0	{"groupName":"71cb224e-bed2-4467-be5f-5de698bfce6d_0","rankedItems":[{"summary":{"itemId":315276240886,"itemVrtId":0

10 result(s) in total

1

* Online Feature Store shared

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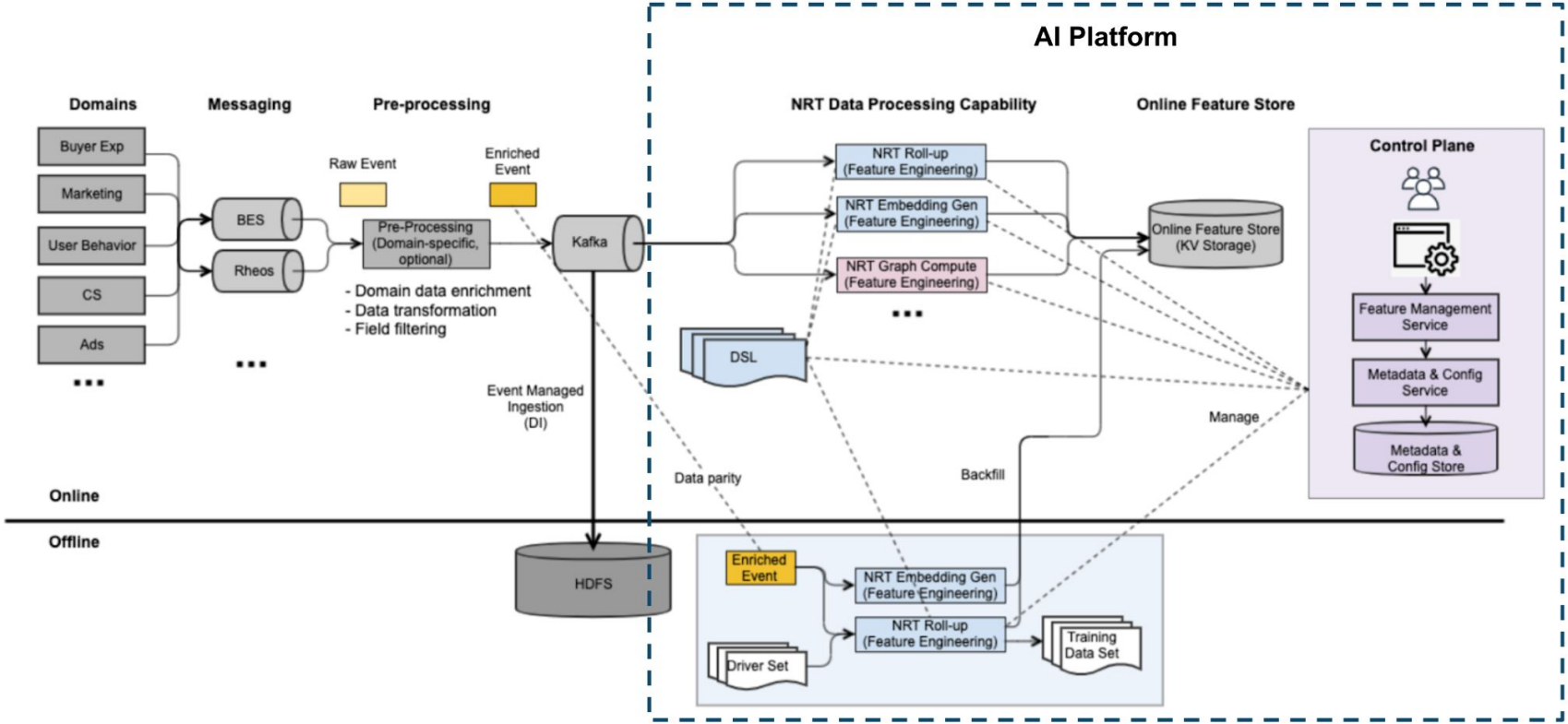
Near Real Time Feature



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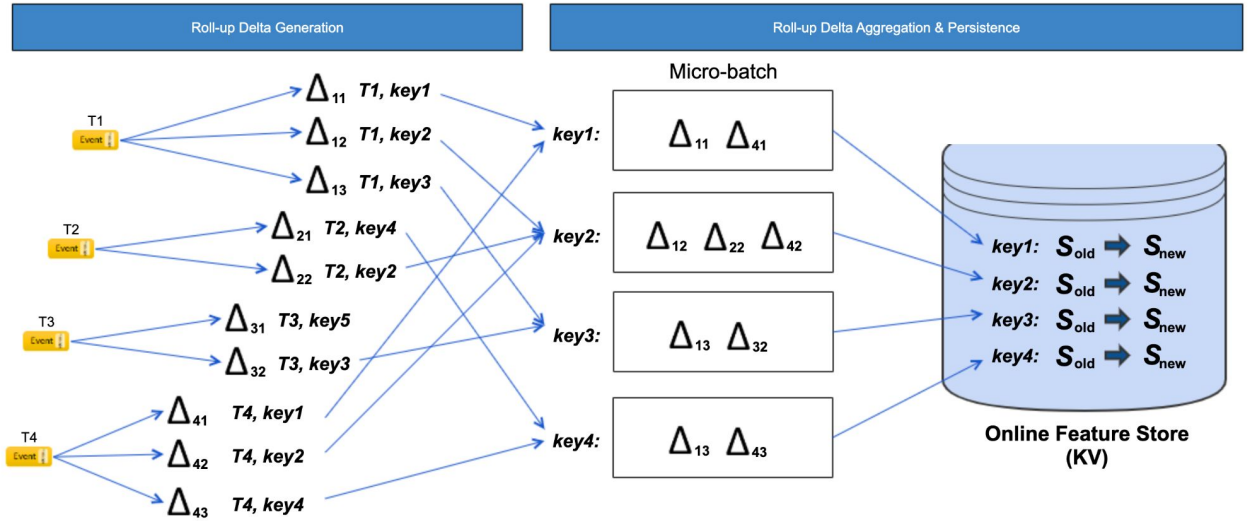
Near Real Time Feature

Generic pipeline with built-in roll-up variable types support

- **Sliding window**
  calculate the counters within some time window
- **LastK**
  Keep last K records and apply logic over
- **Embedding**
  Vector representation of entity's value (image, title, etc)



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Near Real Time Feature Onboarding

Normal | Feature Routing

Type: EVENT
Data Source: AdsE2Eevent
Rollup Key: currentRecord.userId
Read Key: BuyerId
Point-in-Time: currentRecord.eventTimestamp
Variable Type: SLIDING_WINDOW
Aggregation Function: COUNT
Target Fields: currentRecord.clickId
Time Window: 5m

Generate

SLIDING_WINDOW
LAST_K

Search... Columns Filters Density Export

	Alias	Read Key	Data Source	Rollup Key	Type	Function	Window	Default Value	Acti
☑	currentrecord_userid_adse2eevent_currentrec	BuyerId	AdsE2Eevent	currentRecord.us...	NRT	COUNT	5m		

1 result(s) in total

1

Save Features

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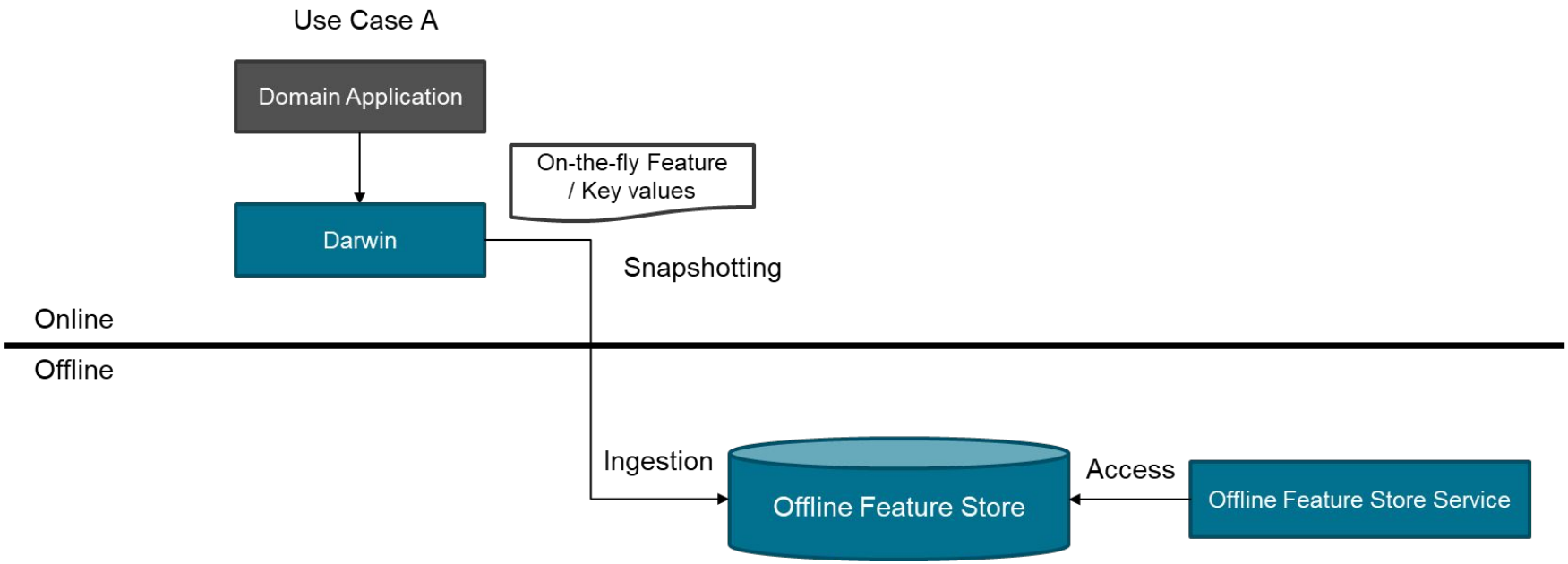
On-the-fly Features & Keys



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Online Serving

- Performance: heart beat/warmup, embedding storage & latency, cache etc.
- Multiple NuKV access

```
{
    "createdTime": null,
    "updatedTime": null,
    "createdBy": "tianlu",
    "updatedBy": null,
    "version": null,
    "labels": null,
    "name": "ads-recommend",
    "schemaId": "v1"                    # schema id will be updated when
    "evalType": "MODEL_SERVING",
    "client": "Darwin",
    "models": [],
    "features": [
        {
            "name": "var3:Account",
            "createdBy": "tianlu",
            "featureType": "NRT",
            "dataType": "INTEGER",
            "readingKey": ["Account"],
            "window": "30m",
            "status": null,
            "featureTemplate": "var3",
            "groupName": "test3",
            "defaultValue": null,
            "wizards": []
        },
        {
            "name": "cnt_merch_user_click_sw^15D:Account",
            "createdBy": "tianlu",
            "featureType": "NRT",
            "dataType": "INTEGER",
            "readingKey": ["Account"],
            "window": "30m",
            "status": null,
            "featureTemplate": "cnt_merch_user_click_sw
```

request payload

```
{
    "requestId": "abc",     # user generated, same id will return in r
    "timeout": 50,          # optional, default no timeout. in millise
    "bestEffort": false,    # optional, default false. if false, any f

    "evalPoint": "ads-recommend",    # EvalPoint name to retrieve feat
    "keys": { "BuyerId": xxx, "SellerId": xxx } # required keys defin
}
```

response

```
# success
{
    "requestId": "abc",

    "schemaId": "v1",                                                   # it tells which
    "results": [
        { "feature": "cnt_merch_user_click_sw^15D:Account",             # feature name i
          "value": xxx,                                                 # computation res
          "success": true or false,                                     # if true, value
          "errorCode": 100                                              # only if success
        },
        { "feature": "cnt_merch_user_click_sw^10D:Account", "value": xxx, "success": true},
    ]
}
```


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Offline Training Set

Feature List

"risk:otf:fg1#connected_bad_slr:Buyer.Id"
"risk:nrt:fg1#sum_lstg_price_by_slr_sw_6h:Seller.Id"
"risk:batch:fg2#trust_mgid_txn_amt:Buyer.Id"

order_id	pit	connected_bad_slr	connected_bad_byr	trust_mgid_txn_cnt	trust_mgid_txn_amt	sum_lstg_price_by_slr_sw_6h	sum_lstg_price_by_slr_sw_24h
03-0867-987247	1653619596	1	4	41	4586.11	0	0
05-0867-866801	1653591001	0	1	8	595.86	555.01	822.44
07-0865-793782	1653168009	1	0	8	402.26	0	0
08-0866-644929	1653359657	6	5	45	795.13	0	2966.3
11-0865-870258	1653229597	1	0	71	6628.23	13.99	365.82
13-0866-133604	1653286060	0	2	31	4696.01	0	1856.28
15-0867-801843	1653659557	1	5	31	780.28	369.2	1677.35
21-0867-403910	1653597472	1	0	28	3475.16	0	0
01-0868-357648	1653691213	1	0	853	7806.73	0	0
08-0865-998310	1653239684	7	9	41	507.86	0	293.22
08-0866-233293	1653286687	1	1	37	1153.93	37692.07	38192.5
09-0867-846933	1653629894	1	0	5	531.1	0	0
10-0865-836963	1653214631	8	1	30	759.58	670.67	1064.47
11-0866-202331	1653296201	1	2	4	334.37	0	0

Driver Set

On-the-fly **Batch Features** **Near Real Time Features**

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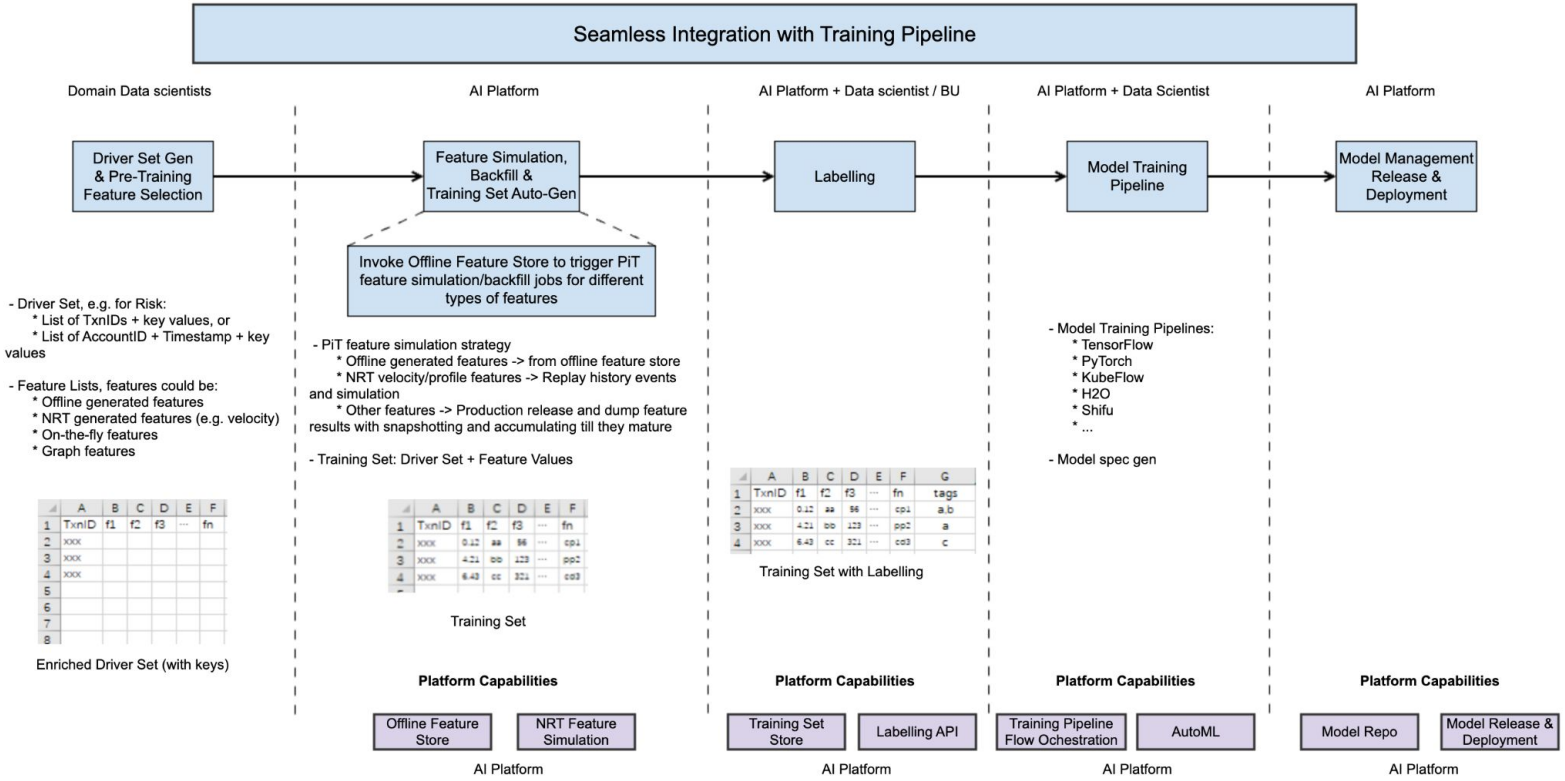
Training Pipeline



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Unified Feature Store

Status

Key Metrics

Single Call Latency < 20ms

7.89 ms

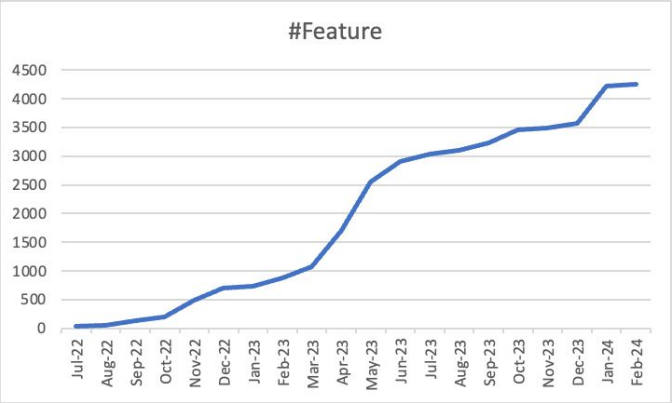
Bulk Call Latency < 50ms

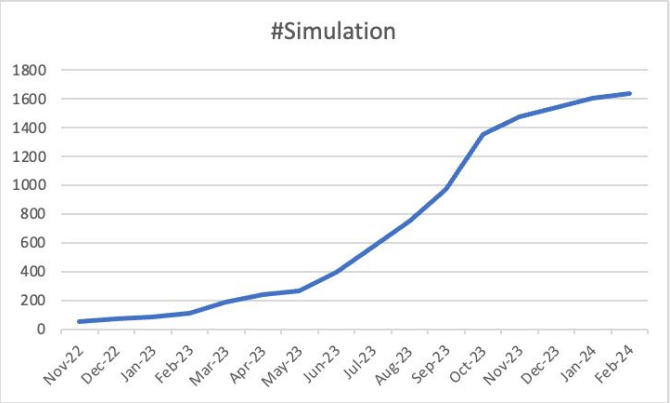
39.8 ms

- **ATB:** ~99.99%
- **NRT feature engineering P99** < 10s
- **Online/Offline match rate:** > 90%
- **Fast feature maturity:** daily

Adoption

Widely adopted by **Risk, Coreai, Recs, Ads, Marketing, BuyerExp, Shipping, DSS, GIS, IE** etc.





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http://go/mlp-sla

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Useful Links

- AI Portal: https://aip.muse.vip.ebay.com/
- User Doc: https://pages.github.corp.ebay.com/taichi/doc-center/docs/feature-store/feature-store-101
- High Level Design: https://wiki.corp.ebay.com/display/TAICHI/Unified+Feature+Store
- NRT Feature Engineering: https://wiki.corp.ebay.com/display/TAICHI/NRT+Feature+Engineering
- API Integration:
  - https://pages.github.corp.ebay.com/taichi/doc-center/docs/feature-store/eval-point
  - https://pages.github.corp.ebay.com/taichi/doc-center/api/featureManagement
  - https://pages.github.corp.ebay.com/taichi/doc-center/api/offlineFeatureStore
- Support Channel: #ml-platform-support

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